Pre & Pro -totyping

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Lesson outline

- Pretotypes
  - The Law of Market Failure
  - Thoughtland vs Actionland
  - Techniques
- Prototypes
  - Fidelity
  - Scope
  - Techniques
- From Wireframes to Prototypes

- Appendix: Prototyping tools

PRETOTYPING

https://www.pretotyping.org/

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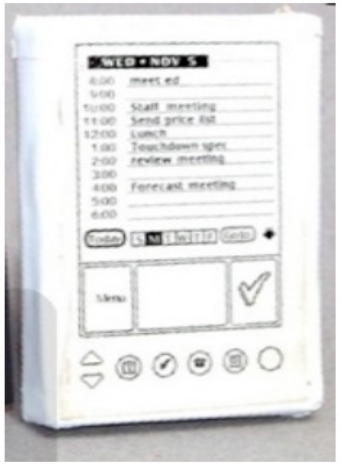
Pretotype It

Make sure you are building the right *it*
before you build *it* right



Alberto Savoia

First Pretotype Edition
August 2011

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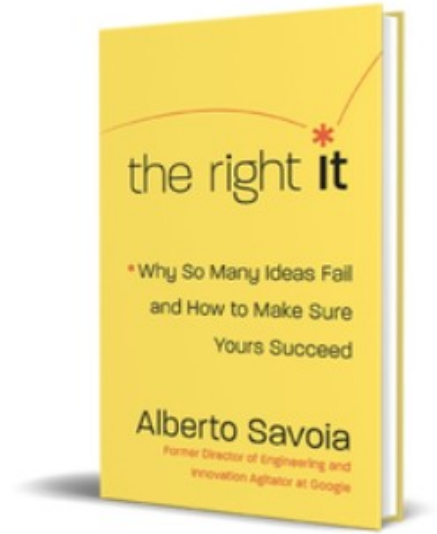

http://www.pretotyping.org/

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Every year, companies launch thousands of new products of all types and in all markets–with each team believing and hoping that "This is the one." All of these launches are carefully followed and tracked by various market research companies. One such company, Nielsen, has been analyzing thousands of worldwide product launches for a long time. Here is a summary from one yearly report:

One year: 24,543 new products*

80%

Failed	27%
Disappointed	16%
Cancelled	37%
Success	14%
Star	6%
Total	**100%**

* Nielsen

Source: www.pretotyping.org

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The law of market failure

- **80-90%** new products fail in the market [1]
- **4/5** startups lose funder's money [2]
- **90%** mobile apps don't produce money
- **78%** project disappear within 4 years
- **16%** to **50%** strategic projects are abandoned
- **47%** companies estimates **50%** of projects fail

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The Law of Market Failure:

Most new products will fail in the market,

even if they are competently executed.

In criminal law, a person is presumed innocent until proven guilty. When it comes to market law, we should presume a potential new product to be a failure—at least until we've collected enough objective evidence to make us believe otherwise.

Source: www.pretotyping.org

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The law of market failure

There is only one way to fight the Law of Market Failure: **test the market** for your ideas objectively, rigorously, and **quickly before you invest to develop them.**

Pretotyping provides you with the tools and techniques you need to validate your idea with minimal resources and in a very short time (as little as a few hours.)

Source: www.pretotyping.org

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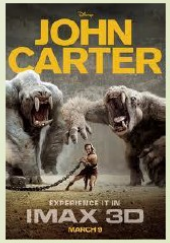
False Positive

- Webvan, late 90s
  - raised $122+M, Goldman Sachs, Sequoia Capital
  - Raised $375M in IPO, 1999
  - In total, raised over $1B
  - Bankruptcy July 2001
- Motorola's Iridium satellite phone system ($6B for 66 satellites)
- Segway (~$180M in funding)
- Google Wave real time messaging platform (~$20-30M)
- "John Carter" by Disney (Cost $275M + $100M in marketing)
- New Coke, Pepsi Clear (est. $50M).



Source: www.pretotyping.org

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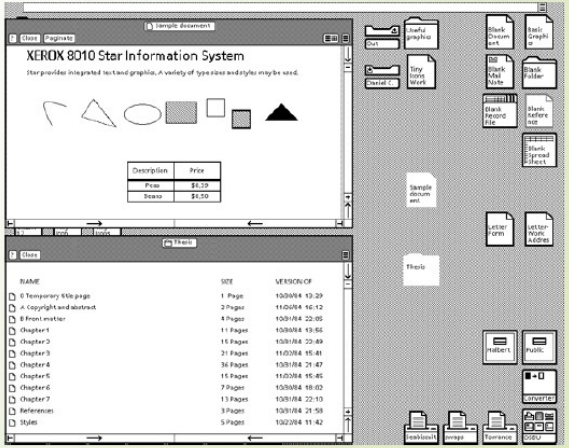
False Negative

- SMS
- Twitter
- Google
- Xerox PARC



Source: www.pretotyping.org

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THOUGHTLAND

The Lost in Translation Problem: An idea is an abstraction–and a subjective one at that; it's something that you imagine or picture in *your* head. The moment you try to communicate what you see in your *mind's eye* to someone else you run into a challenging translation problem–especially if your idea is new and different from anything else they've seen. The way you imagine the new product and its uses may be completely different from the way *they* imagine it.

The Prediction Problem: Even if your audience's abstract understanding of your idea is a close match to your original intention, people are notoriously bad at predicting whether they would actually want or like something they have not yet experienced, or if and how they would actually use it.

Source: www.pretotyping.org

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THOUGHTS WITHOUT DATA ARE JUST OPINIONS

False positives can lead you to believe that your idea is immune to The Law of Market Failure, so you invest too much too soon in a new product that will eventually flop. False negatives, on the other hand, can scare you away from giving your idea a chance, and you end up prematurely scrapping the next Twitter, or Google, or Tesla.

To minimize your chances of getting false positives or negatives you need to collect something more substantial and objective than opinions–especially when the people who give you those opinions have no skin in the game. And the only way to do that is to transport your idea from Thoughtland to a more concrete environment–let's call it *Actionland*.

In **Thoughtland** you use abstract **ideas** to ask hypothetical **questions** and collect **opinions**.

Thoughtland: Ideas -> Questions -> Opinions

In **Actionland** you use **artifacts** to prompt **actions** and collect **data**.

Actionland: Artifacts -> Actions -> Data

Source: www.pretotyping.org

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Pretotype

Prototypes can help you fail faster, but often not fast enough or cheaply enough. The more you invest in something the harder it is to let it go and admit it was the wrong thing. Once you have a "proper prototype" working, it's tempting to work on it a little longer and invest in it a little more: "If we add this one feature I am sure that people will finally use it." Prototypes often turn into *productypes* – a prototype gone too far – and you can kiss fail-fast goodbye.

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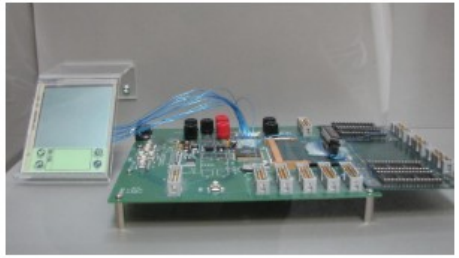
Pretotype

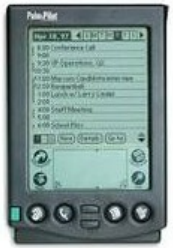
Between abstract ideas and "proper prototypes" there are *pretotypes*. Pretotypes make it possible to collect valuable usage and market data to make a *go/no-go* decision on a new idea at a fraction of the cost of prototypes: hours or days instead of weeks or months, and pennies instead of dollars. Pretotyping helps you fail fast, recover fast and leaves you plenty of time, money, energy and enthusiasm to explore new tweaks or ideas until you hit on something that people seem to want – the rare and wonderful *right it*!

Pretotype

Prototype



Product



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PRETOTYPING FLOW

Step 1: Isolate the Key Assumption: *What is the one assumption about your idea that, if false, means it's definitely not the right it?*

Step 2: Choose a Type of Pretotype: What type of pretotype will let you to isolate and test your key assumption?

Step 3: Make a Market Engagement Hypothesis: How many (and what kind of) people will do what with your pretotype? Your hypothesis can be as simple as: X% of Y will do Z -->A solid hypothesis takes the guesswork and opinion out of testing.
https://www.youtube.com/watch?v=4sZMHAMN0DQ&ab_channel=TheRightIt%E2%80%94VideolessonsbyAlbertoSavoia

Step 4: Test Your Pretotype: Now put your pretotype into the real world, and see how people interact with it. Start small — one place, one time.

Step 5:Learn, Refine, Hypozoom: Evaluate your results. Refine your pretotype with your new data. If you hypothesis held, decide what other situations you should test your pretotype in to get a complete picture (what we call "hypozooming").
https://www.albertosavoia.com/uploads/1/4/0/9/14099067/hypozooming_video.pdf
https://www.youtube.com/watch?v=bKfBbYsJlZc&ab_channel=TheRightIt%E2%80%94VideolessonsbyAlbertoSavoia

Source: www.pretotyping.org

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FAKE DOOR

What A marketing entry point for an as-yet undeveloped idea.

Why The solution doesn't exist yet and you want to capture an initial indication of interest at next to 0 cost.

When Your idea can be concisely described and presented to potential customers where you would expect to find them, and you are confident you can manage the expectations of enthusiastic customers by following up within an appropriate time-frame.

How Advertising a new product or feature, then tracking click-through and customer response rate to see who would be interested in an offering.

Where Web tech enables a very robust method that includes: online ads + landing pages + simple response forms. Same approach also works with emails, offline posters and other media.

the Fake Door can be used to advertise a service that is not ready yet and measure interest from users, e.g. a new process to renew a license, a new expert system to consolidate social services, etc.
https://www.nesta.org.uk/blog/development-and-testing-for-public-labs-fake-it-before-you-make-it/

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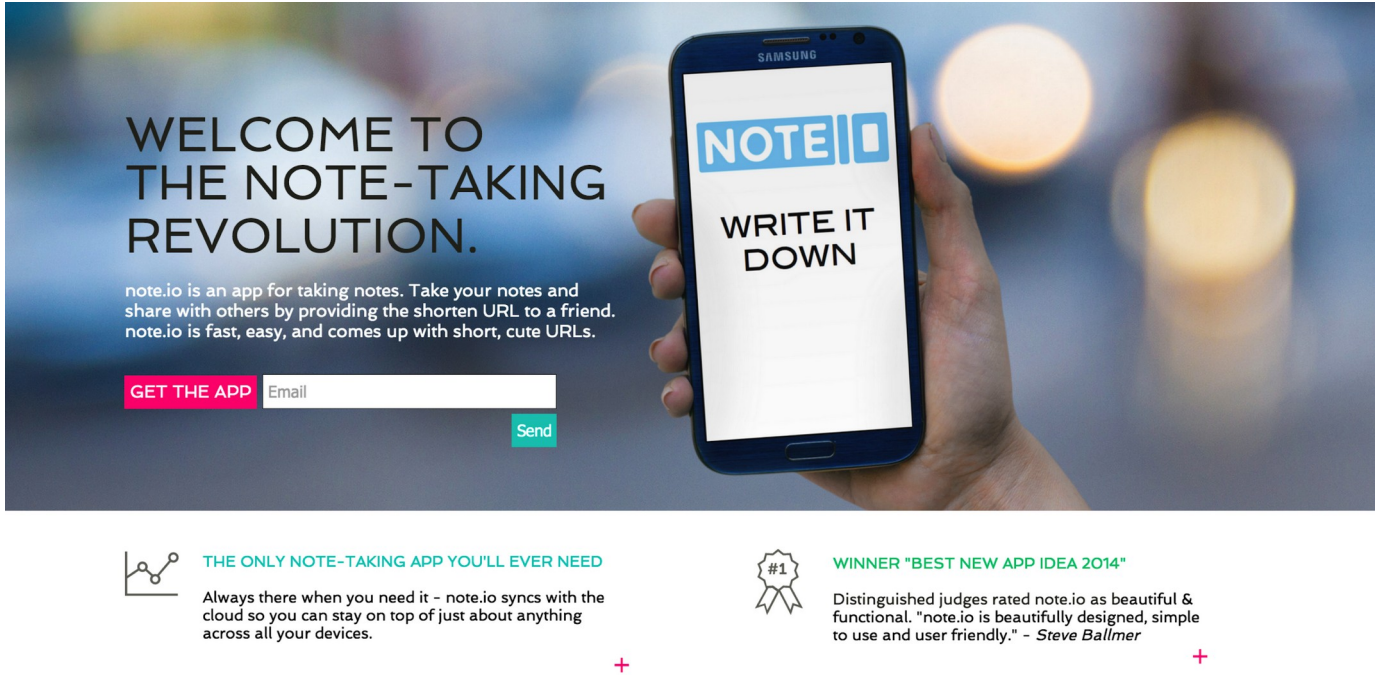

https://hackernoon.com/fake-door-the-mvp-before-the-mvp-61197ed264a3

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MECHANICAL TURK

What	Use human power to simulate a technology that would take too much money or time to build
Why	To postpone costly development until market interest is validated.
When	When the final product requires the development of expensive and complex technology, and those actions and outputs could be simulated by humans.
How	Use a realistic interface to deliver target customers the essential experience of a proposed technology, simulating functionality of a complex back-end using human input.
Where	In the same real-life situation where the innovation will be used.

Algorithmic-based solutions are needed to reach scale while keeping the cost low. But before finding the right "advisor" to build, a Mechanical Turk approach can be used to experiment, where human experts can hide behind an online form or a SMS-based application.
https://www.nesta.org.uk/blog/development-and-testing-for-public-labs-fake-it-before-you-make-it/

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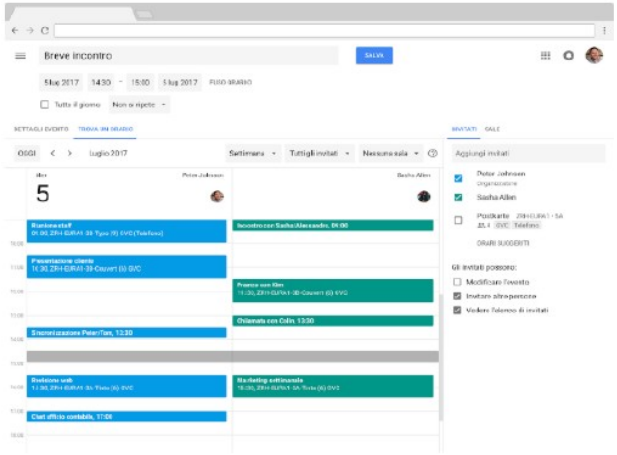
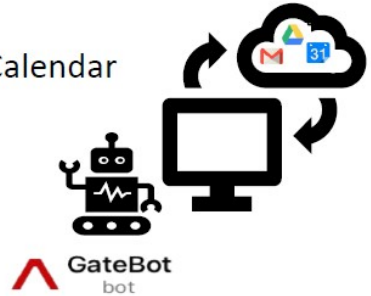
Advanced schedule system



GateBot + G Suite Google Calendar



IMPERSONATOR (The Re-Label)

What	Use an existing product or service to pose as the new offer under test.
Why	Save on development costs while the market interest is not yet validated.
When	A test of the value of the solution depends on the customers' ability to interact with a full-scale design, and you need to create a plausible stand-in for the size, shape, color, features, etc. of the solution.
How	Apply a new skin to an existing product that can act as a good substitute to validate market interest.
Where	In the same real-life situation where the innovation will be used or accessed.

In the context of public labs, this could mean revamping an existing service by providing a different front-end on top of an existing API.
https://www.nesta.org.uk/blog/development-and-testing-for-public-labs-fake-it-before-you-make-it/

Video "IKEA" Design studio Upwell hacks IKEA... on Vimeo

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PINOCCHIO

What	An inanimate (or “dumb”) artifact acts as a proxy for the real thing.
Why	The solution doesn’t exist yet and you want to validate a key design parameter early on.
When	Your solution requires a significant switching or behavioral adaptation by customers to develop a new habit OR You expect demand to be sensitive to the appearance or form factor of your solution.
How	Use a proxy to validate certain parameters of the product like form factor, features and usability.
Where	In the same real-life situation where the innovation will be used.

A Pinocchio pretotype is one in which a fake artifact acts as a proxy for the real thing. The most famous example is the wood model of the Palm Pilot mentioned above.

Pinocchio pretotypes are the perfect conduit for role play design to encourage people to test your product or service, but also to explore other ways that the "it" you are building is tailored to their needs.

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ONE NIGHT STAND

What A complete service experience without the infrastructure required by a permanent solution.

Why Avoid investment in complex infrastructure and validate market interest and actual use.

When
- The solution is — or depends critically upon — an interactive service experience.
- You expect demand for the offer will be sensitive to the choice of channel, and you need to test a number of possible customer interception points.
- You want to validate a large homogeneous market before scaling up.

How Delivering target customers the essential experience within an extremely narrow geo scope and time frame.

Where In the same real-life situation where the innovation will be used but with limited time and geo scope.

The goal here is to reduce cost or to target a well-known population. In the spirit of failing fast, if the One Night Stand pretotype applied to a friendly region and receptive population does not succeed, then you know that you are probably building the wrong "it".

https://www.nesta.org.uk/blog/development-and-testing-for-public-labs-fake-it-before-you-make-it/

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FACADE (The Pretend-to-Own)

What Borrow or rent expensive equipment, space, and assets to simulate a more stable or complex infrastructure underlying your offering.

Why Avoid investments in expensive equipment, space, and other assets while validating interest.

When
- The solution requires major upfront investment, in equipment, space, or assets.
- You expect demand will vary based on customer confidence in your infrastructure.

How Delivering target customers the essential experience, while communicating stability and complexity.

Where In the same real-life situation where the innovation will be used, but with all assets and space borrowed or rented cheaply.

https://www.youtube.com/watch?v=9ko6oFb-jQY

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NEXT STEP… PRODUCTION?

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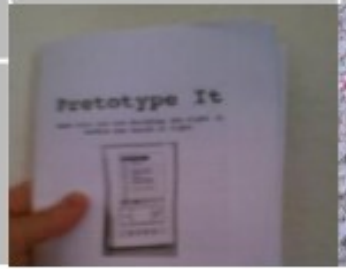
MINIMUM VIABLE PRODUCT

MINIMUM VIABLE PRODUCT	Pretotype it book	An IPod Battery Pack	The Treadmill Bike
WHAT: The transition from pretotyping to prototyping of the eventual product	Is there an audience for this content?	Is it handy to use one?	How is it like to run on a treadmill to propel a bike????
HOW: Creating an artifact which delivers the core function(s) of the full solution			
WHY: you need to put the real product into customers' hands in order to permit a fair test			
WHERE: In the same real-life situation where the innovation will be used			
WHEN: You have learned all you can about market demand from simpler pretotypes (Fake Door, Pinocchio, Mechanical Turk, One Night Stand, or Impersonator) and further insight requires a deeper customer interaction with a functioning artifact			

Source: www.pretotyping.org

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Alberto Savoia Pretotyping Lecture @ Stanford

https://www.youtube.com/watch?v=3sUozPcH4fY&feature=emb_logo

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Prototypes

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What is a Prototype?

A **prototype** is an **experimental** and **early sample** of a design that allows users to visualize and **interact** with it before a final product is developed. Teams build prototypes of varying degrees of fidelity to capture design concepts and test on users.

So, **prototyping** is the **process** in which design teams ideate, experiment with, and **bring concepts to life**. Is the phase in which design teams implement ideas into **tangible** forms from paper to digital.

What is Prototyping? | IxDF (interaction-design.org)

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Why do we need a Prototype?

We never (or at least rarely) get things right the first time!

Prototyping allows designers to **test** their **ideas on users** and **refine** their **designs** based on feedback. This can help save time and resources by identifying potential issues early on.

With prototypes, you can validate your designs so to release the **right product**.

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Why do we need a Prototype?

Prototyping is an essential part of UX design that usually comes after ideation, where you/your team have created and selected ideas that can solve users' needs.

Some advantages of creating a prototype include:

- Providing a solid foundation for ideation and giving stakeholders a clear picture of potential benefits, risks and costs.
- Changes can be made early in the process to avoid costly oversights and commitment to a single version.
- User feedback on the prototype helps identify what works best and if an overhaul is needed.
- Prototyping allows for experimentation with user needs and problems to gain insights into less obvious areas.
- Time-to-market is improved by minimizing errors before product release.

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Prototypes

“They slow us down to speed us up. By taking the time to prototype our ideas, we avoid costly mistakes such as becoming too complex too early and sticking with a weak idea for too long.”

— Tim Brown, CEO & President of IDEO

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Prototypes

In summary:

A prototype is a design solution candidate that addresses specific problems/needs.

With it, usability problems can be identified before committing to the development of the final design.

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How do we do a Prototype?

The prototyping process involves selecting key features to test, creating a design prototype that exemplifies those features, and then testing it on users. There are many different methods for prototyping, ranging from paper sketches to digital versions.

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prototype fidelity, scope and techniques

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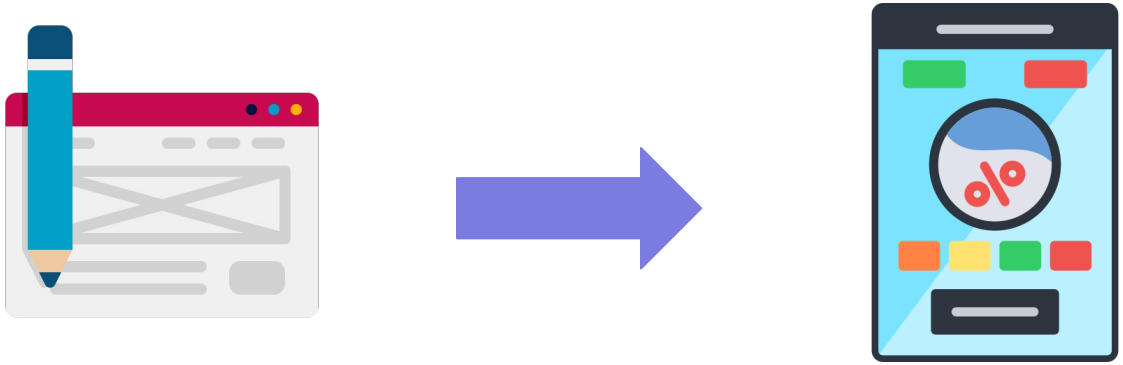
Prototypes fidelity

How much a prototype resembles the final product (in look, feel and function) is a matter of fidelity.

Generally there are 2 types of fidelities (plus another one).



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Low-fidelity prototypes

Pros:

- Usually done with pen & paper (cheap & disposable).
- Connects the low-fidelity wireframes between them.
- Can provide very simple interactions (if not any) and feel.
- Done in less time.
- Easier to modify (more iterations).
- Encourages Design Thinking.

Cons:

- Lack of realism (difficult for users to give feedback).
- Results from early versions may be hard to apply.
- May be too basic to reflect the user experience of the finished product.
- Can oversimplify complex issues.
- Lack of interactivity deprives users of direct control.
- Users must imagine how they would use the product.

https://www.nngroup.com/articles/ux-prototype-hi-lo-fidelity/

https://www.interaction-design.org/literature/topics/prototyping

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Low-fidelity prototypes

https://xd.adobe.com/ideas/process/prototyping/low-fi-and-hi-fi-prototyping/

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High-fidelity prototypes

Pros:

- Can be coded or developed in a specific software (Figma, XD, Sketch).
- Gives a final-like feel.
- You can test more elements (interactions, visual, text readability, etc.).
- Engaging for all stakeholders.
- Allows stakeholders to judge how well the prototype matches users' needs and solves their problems
- Testing yields more accurate and applicable results.

Cons:

- Longer and costlier to create.
- Users may focus on superficial details rather than content.
- Designers may dislike making changes after hours of work.
- Users may mistake the prototype for the finished product and form biases.

https://www.nngroup.com/articles/ux-prototype-hi-lo-fidelity/

https://www.interaction-design.org/literature/topics/prototyping

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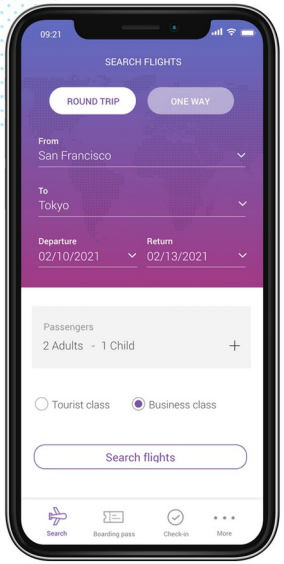
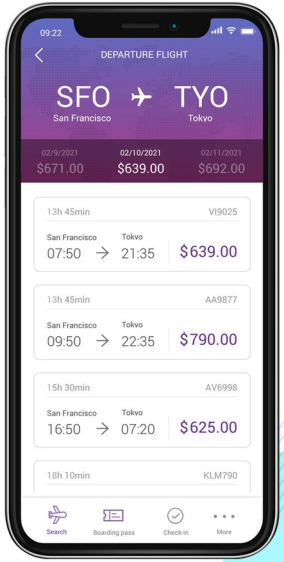
High-fidelity prototypes





https://www.justinmind.com/blog/low-fidelity-vs-high-fidelity-prototypes/

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Finding the right balance: Mid-fidelity prototypes

Another possibility is given by mid-fidelity prototypes where you keep some of the benefits of using Low-fidelity and high-fidelity ones.

An example of a mid-fidelity prototype could be an advanced wireframe with some digital interactivity (i.e., adding interactions to wireflows).

However, you have to remember that the concept of **fidelity is relative**: you have to choose wisely what is best for you depending on the stage of design and features to test.

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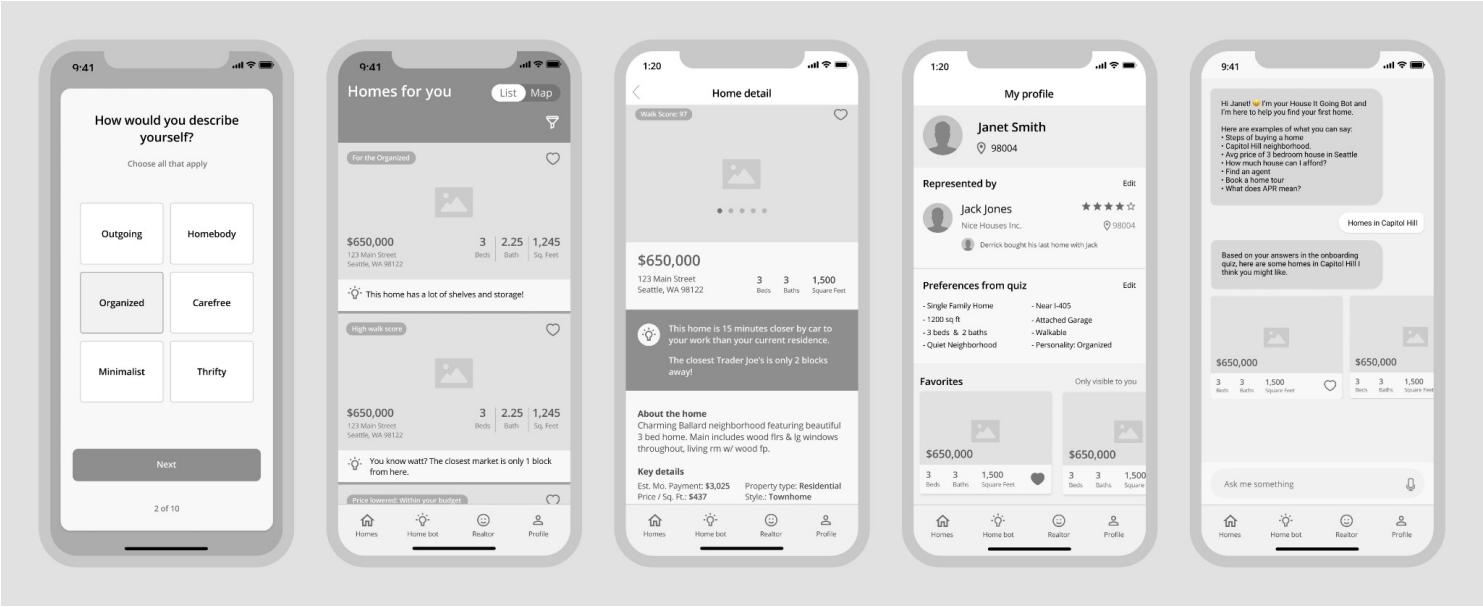
Mid-fidelity prototypes



https://medium.com/the-home-team/homeguide-prototype-portfolio-a8746296e958

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6 dimensions of prototypes' fidelity

The fidelity of a prototype and the activity of prototyping mainly moves on these main 6 dimensions:

1. **Visual/Physical realism**: How closely the prototype resembles the final product in terms of visual design and aesthetics.
2. **Scope**: The breadth and depth of the design represented in the prototype.
3. **Functionality**: What actually works in the prototype, such as links or buttons in a web app.
4. **Data**: Whether the prototype operates on real or fake data.
5. **Autonomy**: Whether the prototype can operate alone or requires input from the designer or user.
6. **Platform**: Whether the prototype is an interim or final implementation.

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Scope of prototypes

Say you have decided to test your design and its functionalities through an mid-fidelity prototype.

This gives you information about how good the prototype will look and feel. **But how much the user will be able to see and test?**

This is a matter of scope of the prototype. There are two possible scopes:

1. Horizontal
2. Vertical

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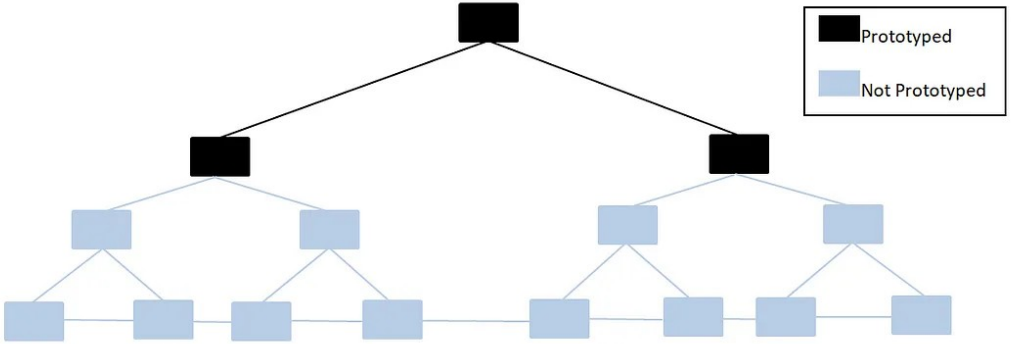
Scope of prototypes: Horizontal

An horizontal prototype gives an entire overview of the system, although not going too much in detail for each feature: you display a wide range of features without actually implementing them. This means that the prototype is constrained to only one layer.

It is useful in the early stages of design as it allows stakeholders to quickly identify potential issues and make design decisions based on feedback from users.



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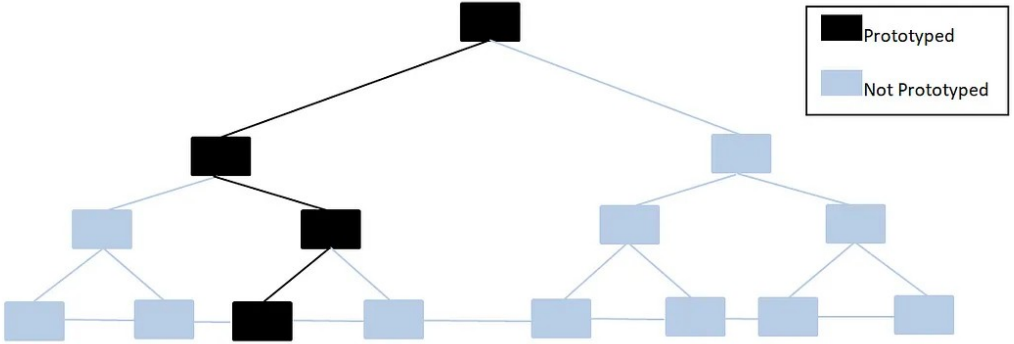
Scope of prototypes: Vertical

A vertical prototype gives a detailed view of a specific feature: you have only one or few features actually designed. This means that the prototype delivers the system and its subsystems.

In this way you can test the technical feasibility of the product or that a specific requirement is fulfilled by the user (in the interaction with the system).



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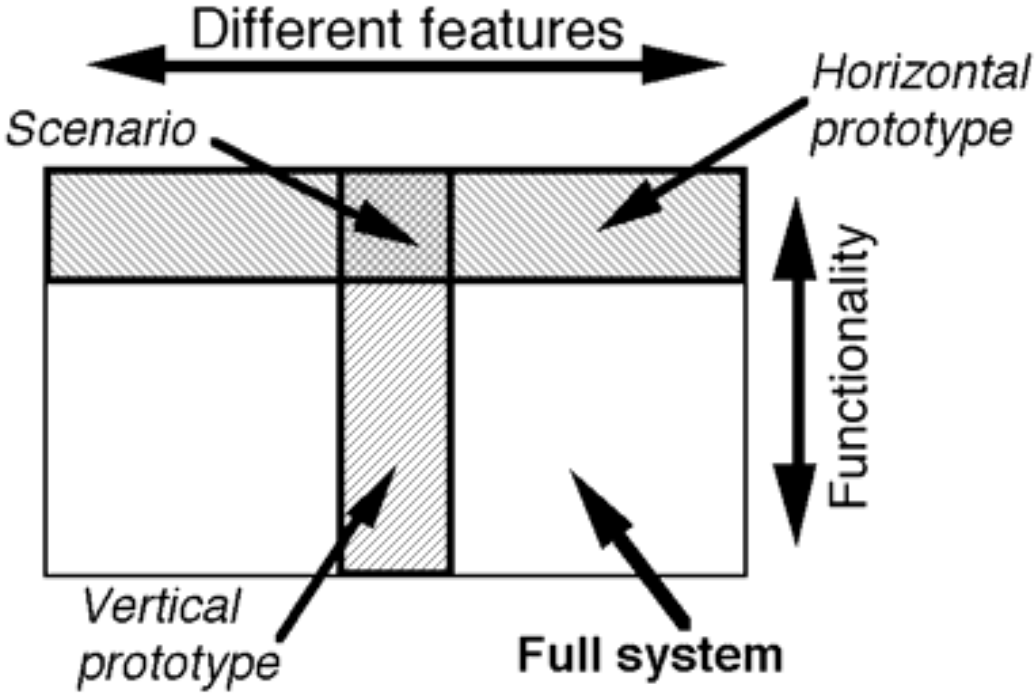
Scope of prototypes



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Prototyping techniques

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Paper prototypes

A paper prototype is a low-fidelity prototype created on paper or cardboard that represents the **basic layout and functionality** of a product or service. It's a **quick and inexpensive** way to test and refine user interface designs before investing time and resources in more high-fidelity prototypes.

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Paper prototypes

Creating a paper prototype involves sketching out the basic layout and functionality of a website or app on paper or cardboard. Users can then interact with the prototype by physically manipulating the paper or cardboard, simulating how they might interact with a digital interface.

Paper prototypes can be used to quickly test a wide range of design elements, from the placement of buttons and menus to the flow of information and user interactions. They are particularly useful in the **early stages** of the design process, when designers are still exploring different design options and gathering feedback from users.

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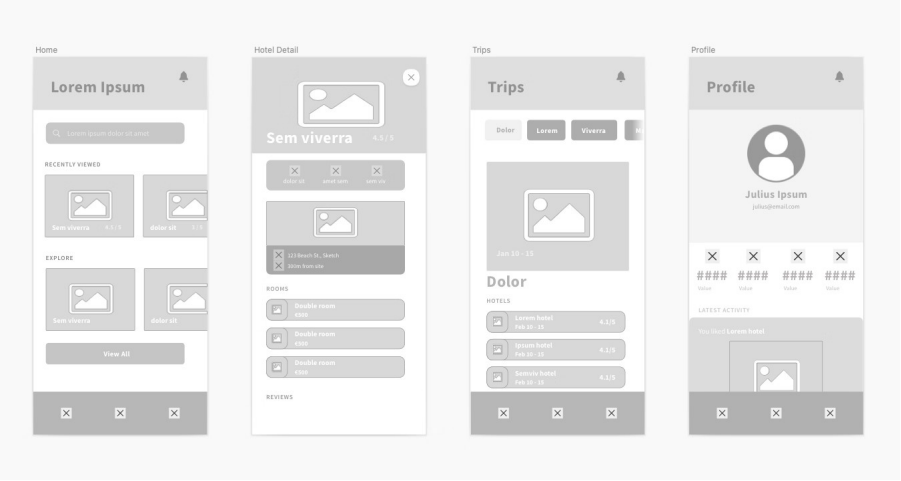
(Mid-fi) Wireframe prototypes

A wireframe prototype is a **digital** prototype that represents the **basic layout and functionality** of a product or service, without much detail or visual design. It's a quick and inexpensive way to test and refine the basic structure of a product or service, without getting bogged down in visual details.



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(Mid-fi) Wireframe prototypes

Creating a wireframe prototype involves sketching out the basic layout and functionality of a product or service, using simple shapes and symbols to represent different UI elements.

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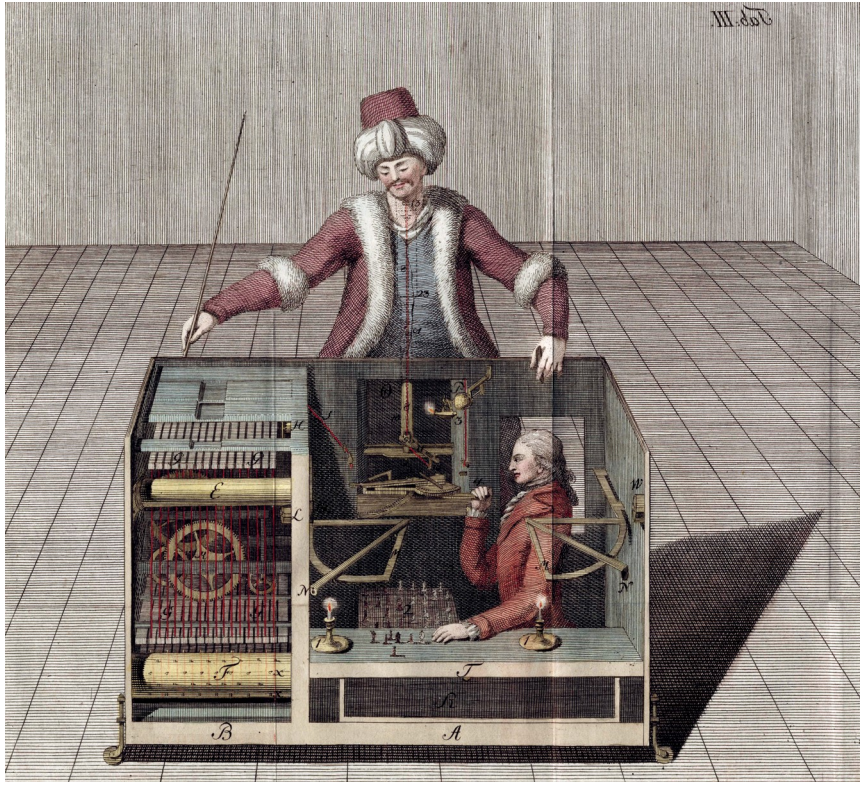
Wizard of Oz (or Mechanical Turk) prototypes

A Wizard of Oz is a prototype that **simulates the behavior** of a product or service **using human input instead of actual technology**. It's used to test the user experience of a product or service before actual technology is developed.



https://en.wikipedia.org/wiki/Mechanical_Turk

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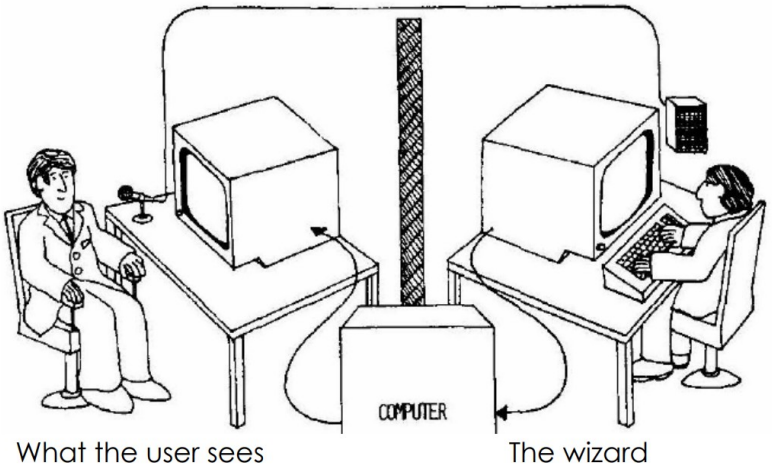
Wizard of Oz prototypes

Creating a Wizard of Oz involves using a human operator or team of operators to simulate the behavior and responses of a product or service. This can involve using pre-recorded audio or video, scripted responses, or even live interactions with users. The goal is to simulate the experience of using a product or service without actually building the technology behind it.



IBM voice editor (1984)

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Functional prototypes

A functional prototype is a prototype that includes **working or partially working features**. It is used to test and refine the **technical aspects** of a product, such as the responsiveness and functionality of different features.

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Functional prototypes

Creating a functional prototype involves building a prototype with actual code and programming, or using a prototyping tool that allows for interactive elements and functionality.

They are particularly useful in the **later stages** of the design process, when designers are looking to fine-tune the technical details of their designs before moving on to full development.

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Non-Functional prototypes

A non-functional prototype is a prototype that looks and feels like a real product or system but **does not have any functional capabilities**. It is typically used to demonstrate the physical design and user interface of a product or system, without the need for any actual working components or systems.

They are typically used to test the design and user experience of a product or system before investing resources into building a functional prototype. A little coding can be still necessary, but without going too much in detail and without real working functionalities or at least just superficially: for example, you have the website and its navigation working but without an actual backend service.

https://www.youtube.com/watch?v=2CTMJxWh5w4

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Non-Functional prototypes

There are several ways that non-functional prototypes can be used for testing:

1. Testing User Experience: can be used to test the user experience (UX) of a product or system. This involves testing how users interact with the prototype, how easy it is for them to navigate and use, and whether it meets their needs and expectations.

2. Testing Visual Design: can also be used to test the visual design as an advanced mockup with some interaction. This involves testing the aesthetics, layout, and overall look and feel of the prototype, and whether it is visually appealing and consistent with the brand identity.

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Wireframes vs Mockups vs Prototypes

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Wireframes vs Mockups vs Prototypes

People often confuse the concept of mockups with that of prototypes. However, prototypes are more advanced versions of mockups that include navigation and functionality, allowing for more meaningful testing and feedback from users.

Designers usually use mockups and wireframes to first create prototypes that simulate the functionality and user experience of the final product.

The prototypes are then used to test and validate the design before moving on to the development phase.

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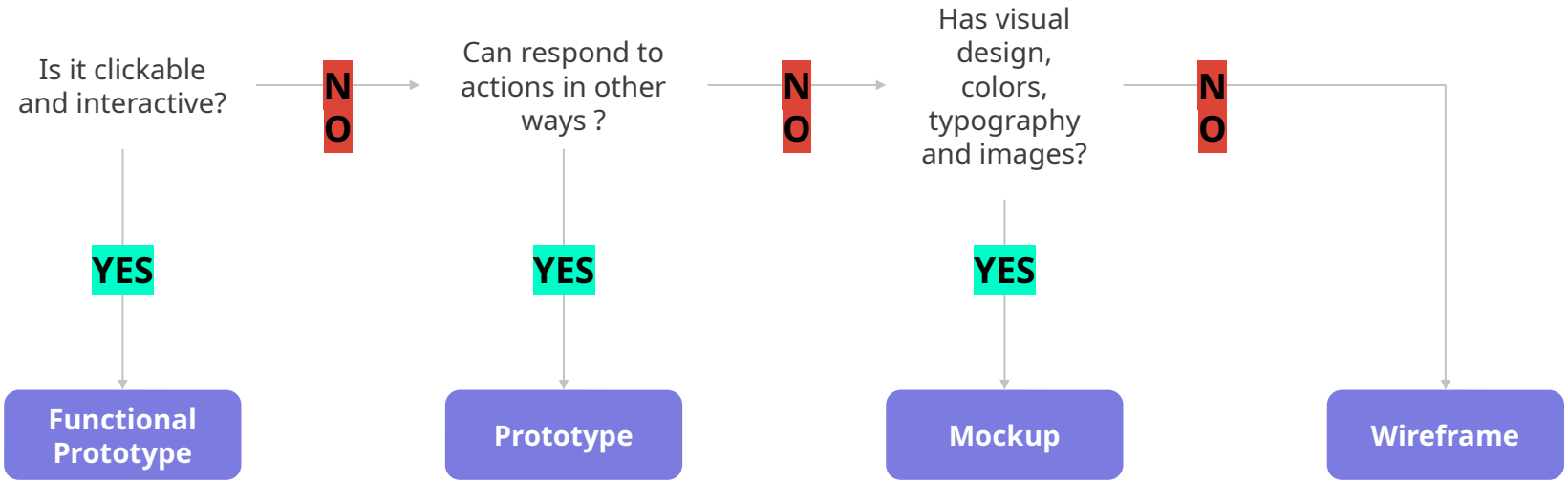
Wireframes vs Mockups vs Prototypes



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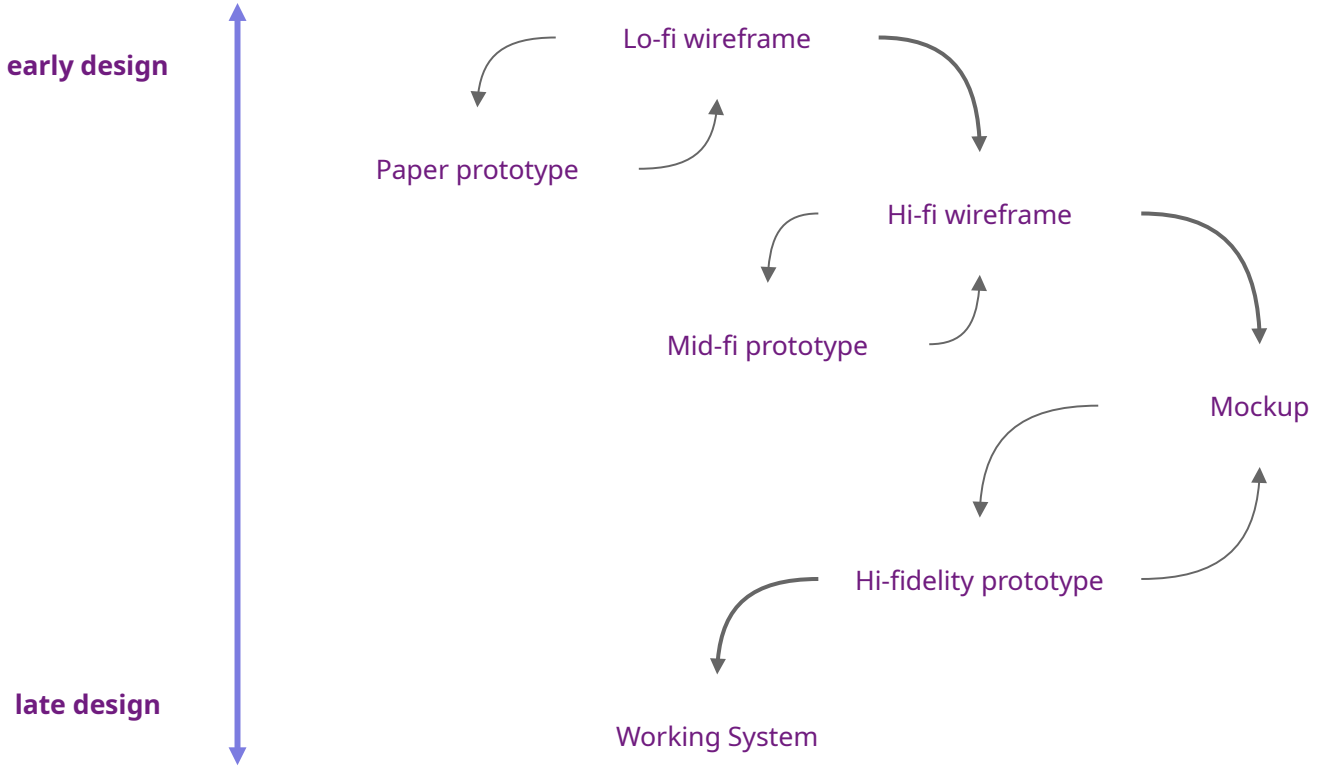
From wireframe to mockups to prototypes



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Prototyping tools (Appendix)

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Tool	Main Features	Platforms	Price
Figma	Collaborative design tool, Vector Networks, Auto Layout, Prototyping, Developer Handoff, Version History	Web-based	Free for individuals (limited features), 12€/month per editor for Professional plan (free for students!)
Sketch	Vector-based design tool, Libraries, Prototyping, Plugins, Handoff	macOS	9€/month per editor
Adobe XD	Vector-based design tool, Auto-Animate, Prototyping, Plugins, Developer Handoff, Voice Prototyping, Collaboration	Windows, macOS	12.19€/month per editor for Single App plan
InVision	Design System Manager, Prototyping, Collaboration, User Testing, Animations, Inspect	Web-based	Free for individuals (limited features), $4.95/month per user for Starter plan (for collaborative teams)

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and others: Axure, Marvel, Principle, Flinto, ProtoPie, Proto.io, UX Pin, ...

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Figma

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Introduction - What is Figma?

Figma is a web app that helps designers build wireframes, mockups, prototypes and so on.

Some advantages:

- The professional plan is free for students.
- Many collaboration features.
- Plugins and community.

https://help.figma.com/hc/en-us/categories/360002051613-Get-started

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Layers on FIgma

Layers work differently on Figma respect Illustrator or other drawing software you may know.

But understanding how to use them is simple:

- Every object in the scene is a layer.
- If you group many objects together they make a layer.
- If you place them in a frame, the frame itself is a layer with many sub-layers.

In addition to this you can use pages to better organize your workplace.

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Some basic but useful shortcuts

When moving objects:

- Shift + mouse move: snap to other objects.
- Alt + mouse move: copy and paste object (on release).
- You can use them together.

When resizing objects:

- Shift + drag borders: resize maintaining aspect ratio.
- Alt + drag borders: resize maintaining center position.
- You can use them together.

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